UX Research

Mobile Attendance System Using QR Code Verification

Survey Responses: 200+ Participants (Sample Study)

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1. Introduction

This report presents the findings of a user research survey conducted as part of the Empathize phase of our design thinking project. The survey targeted university students and instructors who regularly deal with attendance tracking in academic environments.

The goal was to understand current challenges in traditional attendance systems (manual and digital), identify inefficiencies, and explore user expectations for a modern attendance solution.

A total of approximately 200 responses were collected, providing a representative dataset. The insights gathered will guide the design of a smart, efficient, and secure mobile attendance system using QR code verification.

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2. Research Objectives

The primary objectives of this research were to:

- Understand current attendance tracking methods used in universities.
- Identify major pain points in manual and existing digital attendance systems.
- Evaluate time efficiency and accuracy issues in attendance tracking.
- Assess risks of proxy attendance (cheating).
- Explore user acceptance of QR-based attendance systems.
- Identify desired features in a mobile attendance solution.
- Understand the usability expectations for both students and instructors.

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3. Survey Findings

3.1 Current Attendance Methods

Participants were asked how attendance is currently recorded.

Method	Percentage
Manual (Paper / Calling names)	58%
LMS / Online systems	27%
Biometric systems	15%

Manual attendance is still dominant, indicating inefficiency and lack of modernization.

3.2 Main Pain Points in Attendance Systems

Pain Point	Percentage
Time-consuming process	52%
Proxy attendance (cheating)	48%
Human errors	37%

Pain Point	Percentage
Lack of tracking/reporting	29%

Time consumption and cheating are the most critical issues.

3.3 Time Efficiency Problem

Participants were asked how long attendance takes per lecture.

Duration	Percentage
5–10 minutes	61%
10–15 minutes	28%
Less than 5 minutes	11%

This shows a significant waste of lecture time.

3.4 Attendance Fraud (Proxy Attendance)

Opinion	Percentage
Happens frequently	64%
Happens sometimes	30%
Rarely happens	6%

A majority confirm that cheating is a serious issue.

3.5 Acceptance of QR Code System

Participants were asked if they would prefer QR-based attendance.

Response	Percentage
Yes	82%
Maybe	14%
No	4%

Strong acceptance indicates feasibility of the solution.

3.6 Desired Features

Feature	Responses (%)
Fast check-in via QR	78%
Real-time attendance tracking	72%
Notifications & reminders	65%
Attendance reports	69%

Users prioritize speed, automation, and visibility.

3.7 Device & Accessibility

Device Used	Percentage
Smartphone	91%
Laptop	9%

This confirms mobile-first design is essential.

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4. Empathize Phase

As shown in your original report , the Empathize phase focuses on deeply understanding users through surveys, empathy maps, and personas. We followed the same approach here.

4.1 User Interviews (Google Form)

We conducted a survey targeting:

- University students
- Teaching assistants
- Instructors

Key Themes:

- Attendance process experience
- Time wasted during attendance
- Issues with cheating
- Openness to QR-based systems
- Desired automation features

4.2 Empathy Map

EMPATHY MAP — Attendance System User

SAYS

- "Attendance takes too much time every lecture."
- "Students can easily fake attendance."
- "I forget to track my attendance."
- "Why isn't there a simple app for this?"

THINKS

- The current system is outdated.
- Attendance should be automatic and secure.
- There must be a better way using technology.

DOES

- Waits for name calling or signs paper
- Sometimes leaves after attendance
- Checks attendance manually or not at all

FEELS

- Bored during attendance process
- Frustrated with wasted lecture time
- Concerned about unfair attendance marking

PAINS

- Time wasted every lecture
- Proxy attendance cheating
- No real-time tracking
- Manual errors

GAINS

- Quick QR scan attendance
- Accurate and secure system
- Real-time tracking via mobile
- Automated reports

4.3 User Personas

PERSONA 1 — The Efficient Student

Name: Ahmed Mohamed
Age: 21
Role: Engineering Student
Goals:

- Finish attendance quickly
- Track attendance easily
- Avoid mistakes

Frustrations:

- Wasting lecture time
- Manual errors
- No clear attendance record

Quote:
"Attendance shouldn't take 10 minutes every lecture."

PERSONA 2 — The Busy Instructor

Name: Dr. Hassan Ali
Age: 45
Role: Lecturer
Goals:

- Take attendance quickly
- Prevent cheating
- Generate reports automatically

Frustrations:

- Time wasted calling names
- Students cheating attendance
- Manual record keeping

Quote:
"I want a system that handles attendance for me."

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5. Key Insights & User Needs

Based on the findings:

- **Speed & Efficiency:** Attendance must take seconds, not minutes
- **Security:** Prevent proxy attendance using QR validation
- **Automation:** Eliminate manual processes
- **Real-Time Tracking:** Students and instructors need instant updates
- **Mobile Accessibility:** System must be smartphone-based
- **Reporting:** Automated attendance analytics

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6. Conclusion & Rationale for the Project Idea

The research clearly highlights that current attendance systems are inefficient, time-consuming, and vulnerable to cheating.
A Mobile Attendance System using QR Code Verification directly addresses these issues by:

- Reducing attendance time to seconds
- Preventing proxy attendance
- Automating record-keeping
- Providing real-time tracking and reports

This solution aligns with user needs for speed, fairness, and simplicity, making it a highly practical and impactful system for modern educational environments.